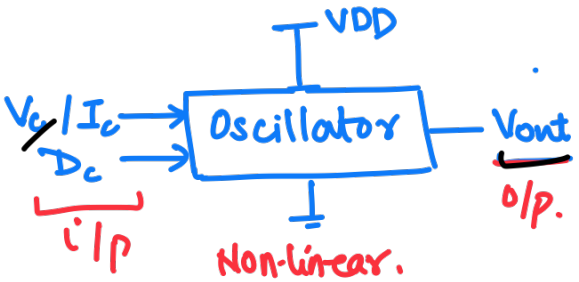
Lecture #36

100p
2p

fin
20MHz



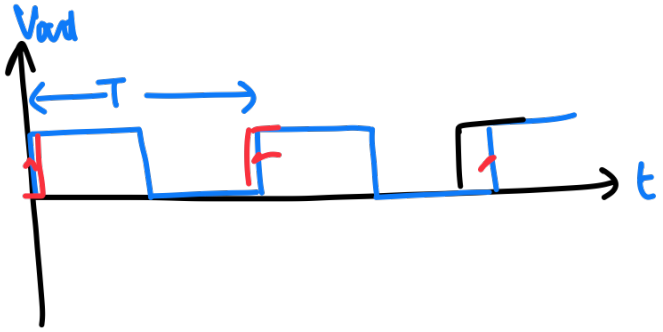
fout = 1GHz.

- V_c: Voltage Controlled Osc. (VCO)
- I_c: Current " " (CCO)
- D_c: Digital " " (DCO)

$$\Delta f_{out} \propto V_c$$

$$V_{out} = f(I_c)$$

$$f_{out} = f(I_c)$$



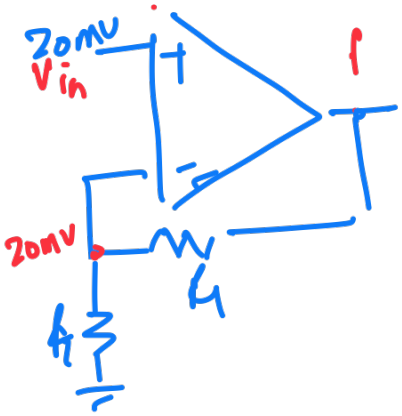
$$f_{out} = \frac{1}{T} = 1GHz$$

Crystal Oscillator are good reference.

$$f_{XTAL} = 20\ MHz.$$

fin = 20MHz ✓

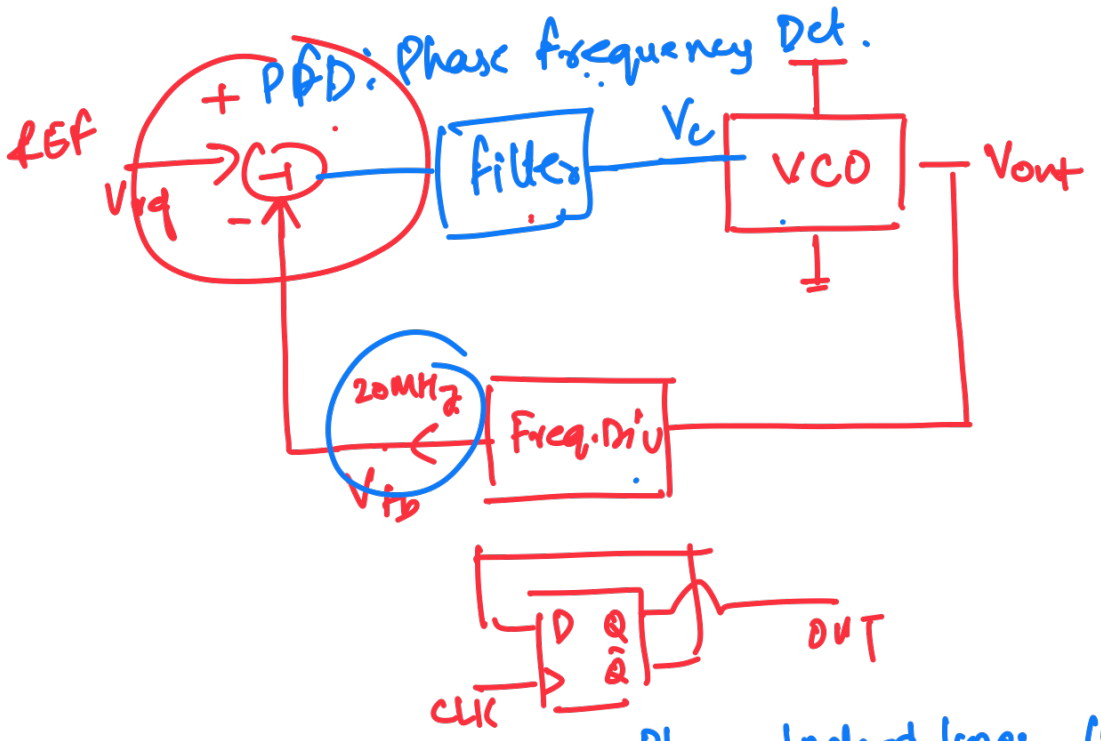
Vin = 20 mV



fout = 1000 MHz.

Vout = 1V

$$f_{out} = \left(1 + \frac{R_1}{R_2}\right) f_{in}.$$



— Phase locked loop < Phase. locked loops (4 credit)
Adv. clk. Gen Tech. (3 credit

— $f_{out} = 50 \times f_{ref}$

$= 50.25\ f_{ref}$.

$\frac{1}{T_{out}} = \frac{50.25}{T_{ref}}$

$T_{ref} = 50.25\ T_{out}$.

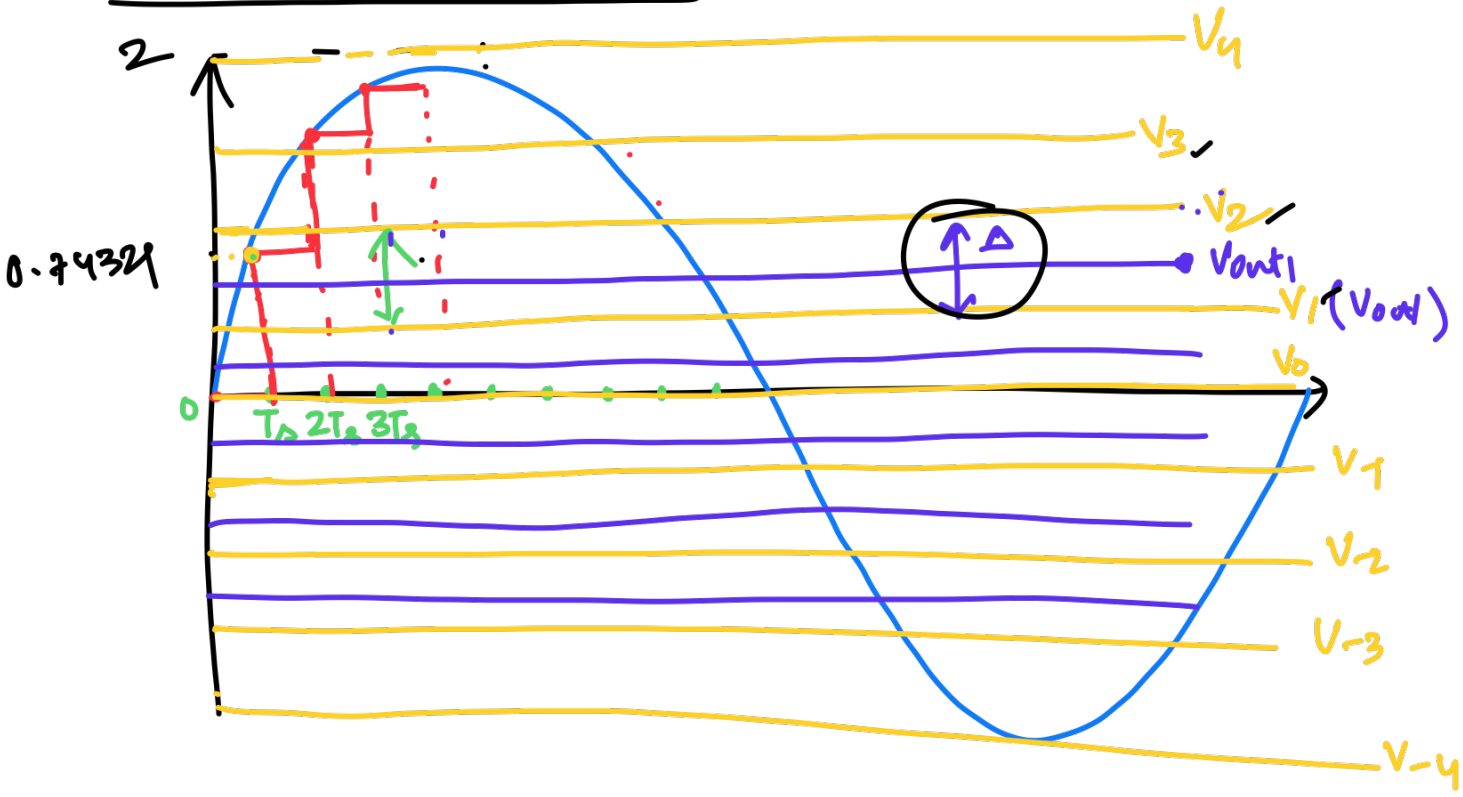
— 1 year = 365 days 6 hrs

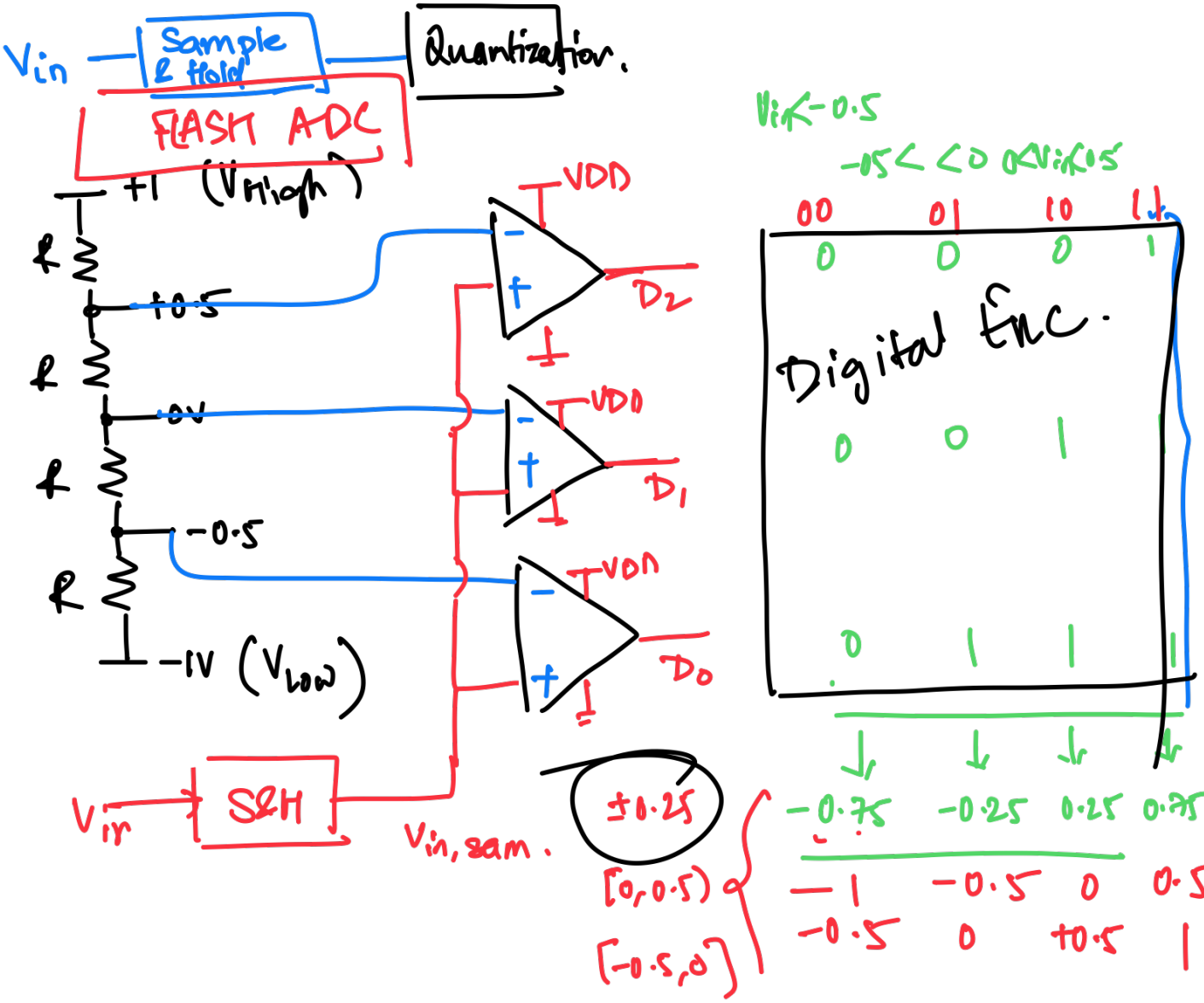
I year = 365
II year = 365
III year = 365
IV year = 366

1 Tref : 50 Tout
2 Tref : 50 Tout
3 Tref : 50 Tout
4 Tref : 51 Tout.

Fractional- N PLL

Analog-to-Digital Conv.





Analog Signal ⟶ 2-bit digital

$2^N - 1$ Comparators.

$$\Delta = \frac{V_{High} - V_{low}}{2^N}$$

N: # of bits used to represent analog signal

- $5'b10111$ (b_4 … b_0)

$$V_{BIN-DEC} = \left(b_0 \cdot 2^0 + b_1 \cdot 2^1 + b_2 2^2 + b_3 2^3 + b_4 2^4 \right) \Delta + V_{off}$$

(4 bits)